CURRICULUM VITAE

Dr. Ghalib H. Alshammri

Associate Professor in AI & Emerging Technologies

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An expert, researcher, and consultant in **Artificial Intelligence**, **Data Science**, **Large Language Models (LLMs)**, and the **Internet of Things (IoT)**, whose work bridges artificial intelligence, governance, energy and sustainability, and healthcare. His research spans bio-molecular nanocommunication networks and the Internet of Bio-Nano Things (IoBNT), agentic AI, and advanced intelligent systems, complemented by extensive consulting experience in strategic government digital transformation projects and organizational AI adoption and implementation initiatives.

✦ Founding & Scientific Leadership

He is the founder of two Saudi scientific societies and the **Founder and Chair of the IEEE Computational Intelligence Society — Saudi Arabia Chapter**. He also serves as a peer reviewer for Q1 journals indexed in the Web of Science. Through his academic, research, and advisory contributions, he advances scientific innovation, responsible AI adoption, and the development of the national artificial intelligence ecosystem in the Kingdom of Saudi Arabia.

20+	40+	12+	9+	3+
Years of academic & professional experience	Research pillars: AI, governance, sustainability, health	Professional certificates from world-class universities	Training programs & bootcamps founded and led	Saudi scientific societies founded, plus an international society

1 Research & Scholarly Impact

A research agenda built on five interlocking pillars — advancing intelligent, responsible, and sustainable technology

🧠 AI & Intelligent Systems

Diffusion-based bio-molecular nano-communication networks and intelligent signal detection — the core of a PhD at Stevens Institute of Technology — extending today into large language models, generative AI, and multi-agent systems.

Nano-networks | LLMs | Multi-Agent AI

⚖️ AI Governance & Responsible AI

Governance, risk, and compliance (GRC) frameworks for AI, data, and cybersecurity; trustworthy and responsible AI adoption; and institutional readiness supporting national digital-transformation and regulatory agendas.

Responsible AI | AI & Data GRC | Policy & Compliance

🌱 Energy & Sustainability

Nanotechnology for energy applications and AI-driven optimization for resource efficiency — advancing green, sustainable, and data-driven smart solutions aligned with Saudi Vision 2030 sustainability goals.

Nanotech for Energy | Green AI | Smart Solutions

🧬 Health & Bio-Intelligence

The Internet of Bio-Nano Things and molecular communications for intra-body healthcare networks; DNA sequence alignment and analysis; and AI applications and education in healthcare settings.

IoBNT | Molecular Comms | AI in Healthcare

📡 Communications & Business

Applying artificial intelligence, data science, and advanced communication technologies to develop smart networks and services, enhance operational efficiency, support strategic decision-making, and innovate business models — strengthening competitiveness and accelerating digital transformation.

Smart Communications · Business Innovation · Digital Transformation

🔍 Q1 Peer Reviewer

Reviewer for first-quartile journals in the Web of Science.

💰 Research & Project Grants

Secured and led grants across academic and government institutions.

👨‍🏫 PhD Supervision

Supervising PhD students and graduate research groups.

🎤 Invited International Talks

Bell Works, NJ (2018) · Honor Global Launch, Dubai (2024).

📚 Publications

Full list on Google Scholar: ksu.sa/tQUC0

2 Work Experience

A career spanning academia, consulting, and institutional founding

Associate Professor in AI and Data Science

Sep 2023 — Present

King Saud University — Riyadh

- Teaching and research supervision in AI, data science, and emerging technologies.
- Supervising graduate students and graduation projects.

Founder and Chairman of the Board

May 2023 — Present

Saudi Society for Artificial Intelligence & Intelligent Systems — Riyadh

- Leading the society and shaping its strategic direction to empower the AI community in Saudi Arabia.

Founder and Chair, IEEE Computational Intelligence Society — Saudi Arabia Chapter

Present

Institute of Electrical and Electronics Engineers (IEEE) — Riyadh

IT, AI and Emerging Technologies Consultant

2021 — Present

Government Entities and Organizations (Projects & Initiatives) — Riyadh

- Specialized consulting for strategic digital-transformation and AI projects and initiatives across the public and private sectors.

Associate Professor (Collaborative / Trainer)

Ongoing collaboration

Several prestigious academic and leadership institutions

- Saudi Electronic University — Riyadh.
- Imam Muhammad Ibn Saud Islamic University — Riyadh.
- King Abdullah College for Leadership and Staff — Riyadh.
- Prince Nayef College for National Security — Riyadh.
- Numerous government and private entities and authorities.

Consultant, Deanship of Scientific Research

Aug 2021 — Jul 2023

Saudi Electronic University — Riyadh

- Advancing the scientific research ecosystem and supporting research and innovation initiatives at the university.

Consultant, Ranking Management — Saudi Global Ranking (Saqr)

Consulting

Education & Training Evaluation Commission — Riyadh

- Supervising the ranking programs platform, surveys, and interactive dashboards.
- Data management and governance for the ranking platform.

Designer, Evaluator and Developer of Standardized Test Questions and Exam Models at Qiyas

Consulting

Education & Training Evaluation Commission — Riyadh

Expert in the AI & IoT Community

Expert

Innovation Center — Monsha'at

Founder and Former Board Member

Dec 2019 — Dec 2021

Saudi Society for Data Science — Riyadh

Collaborating University Professor — Computer Engineering & Electronics Dept.

Mar 2017 — Aug 2020

Stevens Institute of Technology — New Jersey, USA

3 Education

A global academic foundation from the USA, Australia, and Saudi Arabia

🎓 PhD in Computer Engineering (Artificial Intelligence)

Aug 2014 — May 2019

Stevens Institute of Technology — Dept. of Electrical & Computer Engineering, New Jersey, USA

Thesis: Artificial Intelligence Techniques for Diffusion-Based Bio-Molecular Nano Communication Networks. **Research Area:** Molecular communication nanonetworks and AI. **Supervised by:** Prof. Victor B. Lawrence — Associate Dean of Engineering & Batchelor Chair Professor.

🎓 Master of Computer Science (Major: Intelligent Systems)

Feb 2007 — Mar 2008

University of Wollongong — Australia

🎓 Bachelor of Computer Science

Jun 1999 — Jun 2003

King Saud University

GPA: 4.43 / 5 with Second Class Honors — **Ranked First in Class**.

4 Professional Initiatives

Founding and leading distinctive programs with lasting impact

Founding Scientific Societies & International Chapters

• Founder and Chairman of the Board of the **Saudi Society for Artificial Intelligence and Intelligent Systems** — King Saud University.
• Founder and former board member of the **Saudi Society for Data Science** — King Saud University.
• Founder and Chair of the **IEEE Computational Intelligence Society — Saudi Arabia Chapter**.

Designing Academic Programs

Designed three complete academic programs at King Saud University covering **Cloud Computing**, **Cybersecurity**, and **Data Science**.

Interdisciplinary Research Hackathon

Organized the first hackathon dedicated to scientific research and interdisciplinary fields in the Middle East — Saudi Electronic University.

First of its kind in the Middle East

Founding and Delivering Intensive Bootcamps

- **Data Science & Big Data** Bootcamp (60 credit hours) — King Saud University.
- **Business Intelligence** Bootcamp (60 credit hours) — King Saud University.
- **Artificial Intelligence & IoT** Bootcamp (60 credit hours) — King Saud University.
- **Cybersecurity Governance, Risk Management, and Compliance** Program (30 credit hours) — King Saud University.
- **Data Governance, Risk Management, and Compliance** Program (30 credit hours) — King Saud University.

Government and Corporate Training

Designed and delivered a series of specialized technical and engineering training courses for a range of government entities, authorities, and private-sector organizations.

Managing and Organizing the "Riyadh Technical 2022" Forum

Technical and Vocational Training Corporation — held under the auspices of His Royal Highness Prince Faisal bin Bandar bin Abdulaziz Al Saud, Governor of the Riyadh Region. 2022

5 Areas of Expertise

A broad spectrum of technical, research, and leadership specializations

AI Governance & Responsible AI | **AI for Energy & Sustainability** | **AI in Healthcare & Bio-Nano Systems**

Digital Transformation Strategy | Large Language Models (LLMs) | Generative AI | AI Process Automation

AI Agents & Multi-Agent Techniques | Cybersecurity, Data & AI GRC | Internet of Bio-Nano Things

Machine Learning & Deep Learning Applications | Big Data Analytics & Smart Solutions | Academic Curriculum Development

Consulting for Government & Private Sector Projects | Innovation Leadership & Hackathon Management

Data-Driven Decision Making & KPI Design | Training Program Development & Delivery

6 Academic Courses & Supervision

Teaching experience at graduate and undergraduate levels

Stevens Institute of Technology — USA

- Design and Analysis of Network Systems (Graduate)
- Design & Implement IoT with Cloud Systems (Graduate)
- AI Techniques & Big Data (Graduate)
- Supervising research groups for graduate students

Graduate level

King Saud University — Riyadh

- Computer Programming (1) and (2)
- Object-Oriented Programming (OOP)
- Data Structure
- Supervising graduation projects
- Supervising PhD students

Undergraduate & graduate

7 Professional Certifications

Continuous professional development from world-leading universities

Machine Learning
Stanford University

Internet of Things (IoT)
Stanford University — California

Machine Learning for Data Science & Analytics
Columbia University — New York

Quantum Information Science I
Harvard University and MIT

Quantum Machine Learning
University of Toronto

Nanotechnology for Energy
Portland State University — Oregon

DNA Sequences: Alignments and Analysis
University of Maryland

Certified Quality Practitioner
King Saud University

Certified Trainer (Engineering & Technology)
Technical and Vocational Training Corporation — License #8037970980

Academic Leadership
Academic Leadership Center, Ministry of Education, with SACM — Virginia, USA

Key Performance Indicators (KPI) & Benchmarks
King Saud University

Quality Practitioner Preparation & Academic Accreditation
King Saud University

Academic Course Design and Construction
King Saud University

8 Languages & Personal Skills

Leadership and communication skills supporting deep technical expertise

Arabic — Native

English — Advanced proficiency

Leadership | Effective Communication | Self-Motivation | Decision Making | Adaptability | Management Skills

9 Career Highlights

Professional and scientific milestones, locally and internationally

🎤 Speaker at the Honor Global Launch Event — Dubai

Delivered a talk titled "The Future of AI" at the Honor Global Launch Event — Dubai, UAE, 16 July 2024.

🎤 Speaker at Bell Works — New Jersey, USA

Presented a scientific paper on achieving low un-coded bit error rate (BER) for nano molecular communications — 28 November 2018.

🔍 Peer Reviewer for Top-Ranked Journals (Q1)

Serving as a peer reviewer for first-quartile (Q1) journals in the Web of Science, contributing to academic and scientific quality in advanced research fields.

💰 Research & Project Grants

Secured and led multiple research and project grants across academic institutions and government entities, focusing on innovation, digital transformation, and emerging technologies.

🛡️ Strategic Projects

Working on high-impact digital transformation and AI initiatives in collaboration with strategic government and academic entities, involving sensitive data, advanced technologies, and innovative frameworks to enhance organizational efficiency, cybersecurity, and data governance.

🤝 Leadership, Planning & Teamwork

Leading cross-functional teams with a culture of trust and shared vision, excelling in strategic planning and decision-making under pressure, and achieving measurable improvements in effectiveness and productivity in high-demand technical environments.

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